

Orientations

1.1 Orientations of Manifolds

Definition 1.1.1

Let M be a smooth n -manifold with or without boundary, endowed with a pointwise orientations.

- If (E_i) is a local frame of TM , we say that (E_i) is (**positively oriented**) if $(E_1|_p, \dots, E_n|_p)$ is positively oriented basis for T_pM at each point $p \in U$. A **negatively oriented** frame is defined analogously.
- A pointwise orientation is said to be **continuous** if every point of M is in the domain of an oriented local frame.
- An **orientation of M** is a continuous pointwise orientation.
- We say that M is **orientable** if there exists an orientation for it, and **nonorientable** if not.
- An **oriented manifold** is an ordered pair $(M, \mathcal{O})$; an **oriented manifold with boundary** is defined similarly.